Basic Information

Experiment No.: 10

Date: 11-10-2026

Title: File Handling – Counting the Frequency of Patterns "pe" and "pi"

Objective / Aim

To write text into a file named "sample.txt" and count the occurrences of the patterns "pe" and "pi" using Java file handling.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

- Java JDK
- IntelliJ IDEA or any Java IDE
- Operating System: Windows/Linux

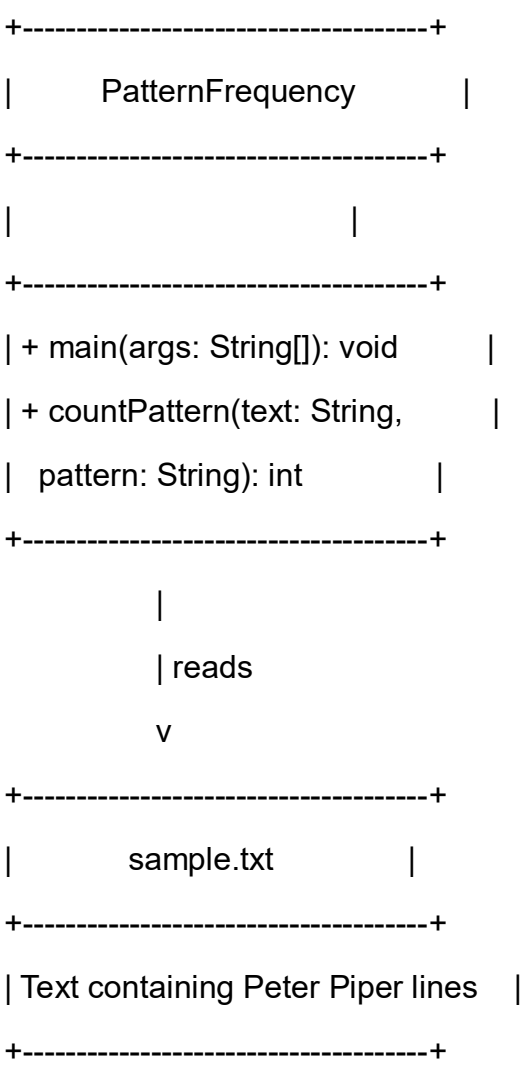
Algorithm / Logic Description

1. Create a file named "sample.txt" containing the given text.
2. Open the file using Java file handling.
3. Read the file contents.
4. Convert the text to lowercase.
5. Count the occurrences of "pe" and "pi".

6. Display the frequency of both patterns.

7. Close the file and end the program.

📐 UML Class Diagram



💻 Source Code

File 1: "sample.txt"

Peter Piper picked a peck of pickled peppers
A peck of pickled peppers Peter Piper picked
If Peter Piper picked a peck of pickled peppers
Where's the peck of pickled peppers Peter Piper picked?

File 2: "PatternFrequency.java"

```
import java.io.File;
import java.io.FileNotFoundException;
import java.util.Scanner;

public class PatternFrequency {

    // Count occurrences of a pattern in the text
    public static int countPattern(String text, String pattern) {
        int count = 0;
        int index = 0;

        // Find each occurrence, including overlapping matches
        while ((index = text.indexOf(pattern, index)) != -1) {
            count++;
            index++;
        }

        return count;
    }

    public static void main(String[] args) {

        StringBuilder text = new StringBuilder();
```

```
        // Read contents from sample.txt
        try (Scanner sc = new Scanner(new File("sample.txt"))) {
            while (sc.hasNextLine()) {
                text.append(sc.nextLine()).append("\n");
            }
        } catch (FileNotFoundException e) {
            System.out.println("sample.txt file not found.");
            return;
        }

        // Convert text to lowercase for case-insensitive counting
        String content = text.toString().toLowerCase();

        // Count the patterns
        int peCount = countPattern(content, "pe");
        int piCount = countPattern(content, "pi");

        // Display the results
        System.out.println("'pe' - no of occurrences - " + peCount);
        System.out.println("'pi' - no of occurrences - " + piCount);
    }
}
```

🔨 Compilation & Execution Commands

Keep "sample.txt" and "PatternFrequency.java" in the same working directory.

Compile

javac PatternFrequency.java

Run

java PatternFrequency

📟 Output Screen

'pe' - no of occurrences - 20

'pi' - no of occurrences - 12

📌 Result

The Java program successfully reads "sample.txt" and counts the occurrences of the patterns "pe" and "pi" using file handling and string operations.